

PROGRESS REPORT 2: Schema Finalization & DDL Implementation.

Course: Database Systems

Group: 3 – Library Management (University Library Management Database)

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1. Introduction.

This project focuses on developing a University Library & Research Management System that models both traditional library operations and academic research management. In addition to borrowing and reservations, the system supports publication management, citation tracking, and textbook-course integration.

The purpose of this report is to document the completion of Phase 2, which involved reviewing and refining the database schema, finalizing the ERD, and implementing the complete DDL script in MySQL. During this phase, we moved beyond basic table creation and focused on ensuring that the schema could properly support real-world workflows such as borrowing, reservations, research access, and citation management.

2. Phase 2 Overview.

The Planned Objective for this phase included:

- Finalizing the ERD diagram
- Write complete SQL schema definitions
- Implement normalization and functionality
- Preparing for query implementation

2.2. During this phase, we completed the following tasks:

- Conducted a detailed review of the initial schema against the five major systems workflows:
- Identified seven structural and logical issues in the original design
- Applied iterative corrections through two revised ERD versions in Miro
- Produced a fully normalized DDL script containing all 16 tables across the three project modules
- Added indexes to improve performance for frequently used queries and workflow operations

Although sample query implementation was initially planned for this phase, we decided to move it to Phase 3 so the queries can be tested using realistic sample data.

3. Major Issues Identified and Corrections carried out.

During our schema review, seven important issues were identified and corrected.

3.1 Redundant Primary Key Structure

The initial design used both generic IDs (id) and business-specific IDs (user_id, book_id) in the same tables. This created unnecessary redundancy, so each table was simplified to use one clearly named primary key.

3.2. Reservation Queue Handling.

The reservation system originally had no method for determining user order in waiting lists. To solve this, a queue_position field was added to support queue-based reservations.

3.3 Missing Academic Information

The textbook_courses table lacked semester and professor information. Both semester and professor_id were added to better reflect real university course structures.

3.4 Research Access Logging

The original schema stored access permissions but did not log actual user activity. A separate research_access_log table was introduced to record access events, timestamps, and actions performed.

3.5. Borrow Limit Support.

A Borrow_limit field was added to the users table to support role-based borrowing restrictions for student, faculty, and staff.

3.6. ENUM Standardization.

Several fields previously used unrestricted VARCHAR values. These were converted into ENUM types to improve consistency and data integrity.

3.7. Additional Review Corrections.

A Second review identified smaller issues involving incorrect foreign key labels and incorrect ENUM values in the research access module. These were corrected before finalization.

4. Finalized Database Schema.

The finalized schema contains 16 tables grouped into three modules and normalized to Third Normal Form (3NF).

4.1 Library Management Module:

- departments
- categories
- users
- books
- Copies
- borrow_records
- reservations
- fines

4.2 Academic Integration Module.

- courses
- textbook_courses

4.3 Research Management Module.

- authors
- publications
- publication_authors
- citations
- research_access
- research_access_log

Key design improvements include:

- Separation of books and physical copies
- Queue-based reservation management
- Role-based borrow limits
- Separation of access permissions from access logs
- Self-referential citation relationships
- Selective use of cascading deletes
- UNIQUE constraints on important business fields

These decisions improved both the realism and maintainability of the system.

5. DDL Implementation.

The final DDL scripts was written for MySQL 8.0+ and includes:

- Database creation with UTF-8 encoding
- All 16 CREATE TABLE statements
- Primary and foreign keys
- CHECK and UNIQUE constraints
- ENUM definitions
- Performance indexes

Special attention was given to reservation queue performance through a composite index on (book_id, status, queue_position). I am currently reviewing the security of this database as well.

6. Challenges Faced.

One of the biggest challenges during this phase was ensuring that the database structure truly matched the workflow diagrams. Some missing fields and relationships only became obvious after comparing the ERD against actual system processes.

Another challenge involved separating research access permissions from access events, which required redesigning part of the schema. We also spent significant time determining appropriate ON DELETE behaviors to balance data integrity with history record preservation.

7. Next Steps:

Phase 3 will focus on implementing realistic sample data and developing SQL queries for the system workflows, including:

- Borrowing and returns
- Reservation queue management
- Research access verification
- Citation Management
- Reporting queries for fines, overdue books, and publication tracking.

Testing and Validation of constraints and workflows will also begin in the next phase.

8. Conclusion.

Phase 2 was completed successfully through multiple rounds of review, correction and refinement. The final database schema now provides a realistic and fully normalized foundation for the University Library Management Database. By correcting structural issues, improving constraints and implementing a complete DDL script, this phase significantly strengthened the overall quality and practicality of the projects.

